Model accuracy analysis

Portable metric summary. Open model_analysis.html for all SVG charts and full details.

model_name: median_baseline
split: calibration
split_role: calibration
n: 11
regression.mae_days: 336.455
regression.rmse_days: 347.03
regression.r2: -15.6614
regression.mean_error_days: 336.455
classification.accuracy: 1
classification.balanced_accuracy: unavailable
classification.macro_precision: 1
classification.macro_recall: 1
classification.macro_f1: 1
classification.precision: unavailable
classification.recall: unavailable
classification.f1: unavailable
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0613152

model_name: median_baseline
split: test
split_role: test
n: 64
regression.mae_days: 351.188
regression.rmse_days: 419.229
regression.r2: -0.0496254
regression.mean_error_days: -91.1562
classification.accuracy: 0.90625
classification.balanced_accuracy: 0.5
classification.macro_precision: 0.453125
classification.macro_recall: 0.5
classification.macro_f1: 0.47541

classification.precision: unavailable
classification.recall: 0
classification.f1: 0
classification.threshold: 0.5
classification.positive_count: 6
classification.auroc: 0.5
classification.average_precision: 0.09375
classification.brier: 0.108637

model_name: median_baseline
split: validation
split_role: tuning
n: 9
regression.mae_days: 263.222
regression.rmse_days: 288.768
regression.r2: -4.9136
regression.mean_error_days: 263.222
classification.accuracy: 1
classification.balanced_accuracy: unavailable
classification.macro_precision: 1
classification.macro_recall: 1
classification.macro_f1: 1
classification.precision: unavailable
classification.recall: unavailable
classification.f1: unavailable
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0613152

model_name: multimodal
split: calibration
split_role: calibration
n: 11
regression.mae_days: 344.386
regression.rmse_days: 388.315
regression.r2: -19.8615
regression.mean_error_days: 344.386

classification.accuracy: 0.909091
classification.balanced_accuracy: unavailable
classification.macro_precision: 0.5
classification.macro_recall: 0.454545
classification.macro_f1: 0.47619
classification.precision: 0
classification.recall: unavailable
classification.f1: 0
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0358912

model_name: multimodal
split: test
split_role: test
n: 64
regression.mae_days: 332.998
regression.rmse_days: 399.283
regression.r2: 0.047877
regression.mean_error_days: 122.575
classification.accuracy: 0.890625
classification.balanced_accuracy: 0.640805
classification.macro_precision: 0.666102
classification.macro_recall: 0.640805
classification.macro_f1: 0.651904
classification.precision: 0.4
classification.recall: 0.333333
classification.f1: 0.363636
classification.threshold: 0.5
classification.positive_count: 6
classification.auroc: 0.761494
classification.average_precision: 0.287701
classification.brier: 0.0934136

model_name: multimodal
split: validation
split_role: tuning

n: 9
regression.mae_days: 426.986
regression.rmse_days: 458.544
regression.r2: -13.9114
regression.mean_error_days: 426.986
classification.accuracy: 1
classification.balanced_accuracy: unavailable
classification.macro_precision: 1
classification.macro_recall: 1
classification.macro_f1: 1
classification.precision: unavailable
classification.recall: unavailable
classification.f1: unavailable
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0319264

model_name: multimodal_late_fusion
split: calibration
split_role: calibration
n: 11
regression.mae_days: 423.186
regression.rmse_days: 452.023
regression.r2: -27.2682
regression.mean_error_days: 423.186
classification.accuracy: 1
classification.balanced_accuracy: unavailable
classification.macro_precision: 1
classification.macro_recall: 1
classification.macro_f1: 1
classification.precision: unavailable
classification.recall: unavailable
classification.f1: unavailable
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable

classification.average_precision: unavailable
classification.brier: 0.0192577

model_name: multimodal_late_fusion
split: test
split_role: test
n: 64
regression.mae_days: 317.13
regression.rmse_days: 374.878
regression.r2: 0.160712
regression.mean_error_days: 146.704
classification.accuracy: 0.890625
classification.balanced_accuracy: 0.491379
classification.macro_precision: 0.452381
classification.macro_recall: 0.491379
classification.macro_f1: 0.471074
classification.precision: 0
classification.recall: 0
classification.f1: 0
classification.threshold: 0.5
classification.positive_count: 6
classification.auroc: 0.755747
classification.average_precision: 0.262446
classification.brier: 0.0877057

model_name: multimodal_late_fusion
split: validation
split_role: tuning
n: 9
regression.mae_days: 449.366
regression.rmse_days: 466.315
regression.r2: -14.4211
regression.mean_error_days: 449.366
classification.accuracy: 1
classification.balanced_accuracy: unavailable
classification.macro_precision: 1
classification.macro_recall: 1
classification.macro_f1: 1
classification.precision: unavailable

classification.recall: unavailable
classification.f1: unavailable
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0302069

model_name: structured
split: calibration
split_role: calibration
n: 11
regression.mae_days: 546.22
regression.rmse_days: 551.839
regression.r2: -41.1311
regression.mean_error_days: 546.22
classification.accuracy: 1
classification.balanced_accuracy: unavailable
classification.macro_precision: 1
classification.macro_recall: 1
classification.macro_f1: 1
classification.precision: unavailable
classification.recall: unavailable
classification.f1: unavailable
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0517567

model_name: structured
split: test
split_role: test
n: 64
regression.mae_days: 372.677
regression.rmse_days: 432.384
regression.r2: -0.116528
regression.mean_error_days: 161.87
classification.accuracy: 0.890625

classification.balanced_accuracy: 0.491379
classification.macro_precision: 0.452381
classification.macro_recall: 0.491379
classification.macro_f1: 0.471074
classification.precision: 0
classification.recall: 0
classification.f1: 0
classification.threshold: 0.5
classification.positive_count: 6
classification.auroc: 0.612069
classification.average_precision: 0.15463
classification.brier: 0.119593

model_name: structured
split: validation
split_role: tuning
n: 9
regression.mae_days: 522.137
regression.rmse_days: 535.119
regression.r2: -19.3074
regression.mean_error_days: 522.137
classification.accuracy: 1
classification.balanced_accuracy: unavailable
classification.macro_precision: 1
classification.macro_recall: 1
classification.macro_f1: 1
classification.precision: unavailable
classification.recall: unavailable
classification.f1: unavailable
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0715323

model_name: structured_extra_trees
split: calibration
split_role: calibration
n: 11

regression.mae_days: 512.177
regression.rmse_days: 520.93
regression.r2: -36.5437
regression.mean_error_days: 512.177
classification.accuracy: 1
classification.balanced_accuracy: unavailable
classification.macro_precision: 1
classification.macro_recall: 1
classification.macro_f1: 1
classification.precision: unavailable
classification.recall: unavailable
classification.f1: unavailable
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0575127

model_name: structured_extra_trees
split: test
split_role: test
n: 64
regression.mae_days: 354.218
regression.rmse_days: 418.79
regression.r2: -0.0474283
regression.mean_error_days: 167.48
classification.accuracy: 0.90625
classification.balanced_accuracy: 0.5
classification.macro_precision: 0.453125
classification.macro_recall: 0.5
classification.macro_f1: 0.47541
classification.precision: unavailable
classification.recall: 0
classification.f1: 0
classification.threshold: 0.5
classification.positive_count: 6
classification.auroc: 0.670977
classification.average_precision: 0.194907

classification.brier: 0.108817

model_name: structured_extra_trees
split: validation
split_role: tuning
n: 9
regression.mae_days: 501.264
regression.rmse_days: 520.323
regression.r2: -18.2
regression.mean_error_days: 501.264
classification.accuracy: 1
classification.balanced_accuracy: unavailable
classification.macro_precision: 1
classification.macro_recall: 1
classification.macro_f1: 1
classification.precision: unavailable
classification.recall: unavailable
classification.f1: unavailable
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0652022

model_name: structured_gradient_boosting
split: calibration
split_role: calibration
n: 11
regression.mae_days: 500.201
regression.rmse_days: 606.183
regression.r2: -49.8377
regression.mean_error_days: 497.031
classification.accuracy: 0.909091
classification.balanced_accuracy: unavailable
classification.macro_precision: 0.5
classification.macro_recall: 0.454545
classification.macro_f1: 0.47619
classification.precision: 0
classification.recall: unavailable

classification.f1: 0
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.066409

model_name: structured_gradient_boosting
split: test
split_role: test
n: 64
regression.mae_days: 372.389
regression.rmse_days: 491.099
regression.r2: -0.440352
regression.mean_error_days: 191.905
classification.accuracy: 0.78125
classification.balanced_accuracy: 0.431034
classification.macro_precision: 0.446429
classification.macro_recall: 0.431034
classification.macro_f1: 0.438596
classification.precision: 0
classification.recall: 0
classification.f1: 0
classification.threshold: 0.5
classification.positive_count: 6
classification.auroc: 0.543103
classification.average_precision: 0.119539
classification.brier: 0.134106

model_name: structured_gradient_boosting
split: validation
split_role: tuning
n: 9
regression.mae_days: 526.019
regression.rmse_days: 643.233
regression.r2: -28.3421
regression.mean_error_days: 526.019
classification.accuracy: 0.777778
classification.balanced_accuracy: unavailable

classification.macro_precision: 0.5
classification.macro_recall: 0.388889
classification.macro_f1: 0.4375
classification.precision: 0
classification.recall: unavailable
classification.f1: 0
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.124767

model_name: structured_hist_gradient_boosting
split: calibration
split_role: calibration
n: 11
regression.mae_days: 562.333
regression.rmse_days: 677.433
regression.r2: -62.4907
regression.mean_error_days: 528.964
classification.accuracy: 0.909091
classification.balanced_accuracy: unavailable
classification.macro_precision: 0.5
classification.macro_recall: 0.454545
classification.macro_f1: 0.47619
classification.precision: 0
classification.recall: unavailable
classification.f1: 0
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0521716

model_name: structured_hist_gradient_boosting
split: test
split_role: test
n: 64
regression.mae_days: 447.139

regression.rmse_days: 589.993
regression.r2: -1.07885
regression.mean_error_days: 249.917
classification.accuracy: 0.78125
classification.balanced_accuracy: 0.505747
classification.macro_precision: 0.503704
classification.macro_recall: 0.505747
classification.macro_f1: 0.5
classification.precision: 0.1
classification.recall: 0.166667
classification.f1: 0.125
classification.threshold: 0.5
classification.positive_count: 6
classification.auroc: 0.574713
classification.average_precision: 0.121775
classification.brier: 0.162656

model_name: structured_hist_gradient_boosting
split: validation
split_role: tuning
n: 9
regression.mae_days: 534.394
regression.rmse_days: 647.589
regression.r2: -28.7409
regression.mean_error_days: 534.394
classification.accuracy: 0.777778
classification.balanced_accuracy: unavailable
classification.macro_precision: 0.5
classification.macro_recall: 0.388889
classification.macro_f1: 0.4375
classification.precision: 0
classification.recall: unavailable
classification.f1: 0
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.103201

model_name: structured_random_forest
split: calibration
split_role: calibration
n: 11
regression.mae_days: 420.869
regression.rmse_days: 451.951
regression.r2: -27.2592
regression.mean_error_days: 420.869
classification.accuracy: 1
classification.balanced_accuracy: unavailable
classification.macro_precision: 1
classification.macro_recall: 1
classification.macro_f1: 1
classification.precision: unavailable
classification.recall: unavailable
classification.f1: unavailable
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0501833

model_name: structured_random_forest
split: test
split_role: test
n: 64
regression.mae_days: 350.55
regression.rmse_days: 428.373
regression.r2: -0.0959119
regression.mean_error_days: 166.239
classification.accuracy: 0.859375
classification.balanced_accuracy: 0.474138
classification.macro_precision: 0.45082
classification.macro_recall: 0.474138
classification.macro_f1: 0.462185
classification.precision: 0
classification.recall: 0
classification.f1: 0

classification.threshold: 0.5
classification.positive_count: 6
classification.auroc: 0.672414
classification.average_precision: 0.157863
classification.brier: 0.112787

model_name: structured_random_forest
split: validation
split_role: tuning
n: 9
regression.mae_days: 511.651
regression.rmse_days: 543.226
regression.r2: -19.9274
regression.mean_error_days: 511.651
classification.accuracy: 1
classification.balanced_accuracy: unavailable
classification.macro_precision: 1
classification.macro_recall: 1
classification.macro_f1: 1
classification.precision: unavailable
classification.recall: unavailable
classification.f1: unavailable
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0742645

model_name: text
split: calibration
split_role: calibration
n: 11
regression.mae_days: 387.926
regression.rmse_days: 415.887
regression.r2: -22.9292
regression.mean_error_days: 387.926
classification.accuracy: 1
classification.balanced_accuracy: unavailable
classification.macro_precision: 1

classification.macro_recall: 1
classification.macro_f1: 1
classification.precision: unavailable
classification.recall: unavailable
classification.f1: unavailable
classification.threshold: 0.5
classification.positive_count: 0
classification.auroc: unavailable
classification.average_precision: unavailable
classification.brier: 0.0161505

model_name: text
split: test
split_role: test
n: 64
regression.mae_days: 314.701
regression.rmse_days: 366.982
regression.r2: 0.195694
regression.mean_error_days: 112.299
classification.accuracy: 0.90625
classification.balanced_accuracy: 0.5
classification.macro_precision: 0.453125
classification.macro_recall: 0.5
classification.macro_f1: 0.47541
classification.precision: unavailable
classification.recall: 0
classification.f1: 0
classification.threshold: 0.5
classification.positive_count: 6
classification.auroc: 0.807471
classification.average_precision: 0.348118
classification.brier: 0.0775694

model_name: text
split: validation
split_role: tuning
n: 9
regression.mae_days: 421.023
regression.rmse_days: 445.862

regression.r2: -13.0979

regression.mean_error_days: 421.023

classification.accuracy: 1

classification.balanced_accuracy: unavailable

classification.macro_precision: 1

classification.macro_recall: 1

classification.macro_f1: 1

classification.precision: unavailable

classification.recall: unavailable

classification.f1: unavailable

classification.threshold: 0.5

classification.positive_count: 0

classification.auroc: unavailable

classification.average_precision: unavailable

classification.brier: 0.0196582

Retrospective descriptive research decision support; no clinical validation or guaranteed future accuracy.

Tuning data support model selection; calibration data fit transforms/intervals; only test data provide the untouched held-out estimate. Test inspection can influence subsequent experiments.

Metrics are recomputed from every stored prediction. Saved aggregate metrics are parsed and hashed but are not treated as evidence of accuracy.

Current registry snapshots may contain post-start updates and leakage. Completed-only and convenience samples can create selection bias; chronological splits do not establish prospective validity.

No regression accuracy percentage is defined. Duration errors use days. Binary decisions use probability >= 0.5; confusion matrices have actual rows and predicted columns.

Undefined values are null, displayed as unavailable. One-class splits do not identify class discrimination. Small held-out samples and imbalance can make estimates unstable; no uncertainty interval is invented.

Reliability uses ten equal-width bins. Raw and post-hoc calibrated probabilities are analyzed separately; calibration-cohort diagnostics can be optimistic.

Duration-by-phase subgroup metrics cannot be recomputed because saved duration prediction rows do not contain verified phase labels; stored subgroup aggregates are not substituted.

Charts show all stored points (overlapping points can obscure sample counts); metrics and bin counts include every row.

Tuning validation metrics are optimistic after hyperparameter and raw model selection; test metrics are held out. Calibration diagnostics are in-sample for the calibration transform.

Bootstrap intervals are conditional on the fitted models and assume independent trials; they exclude model-fitting uncertainty and sponsor/site clustering.

Primary probabilities and classification metrics remain raw. Separate sigmoid-calibrated probabilities/metrics exist only with at least 10 calibration rows and two labels of each class; calibration never selects models.

Split-conformal nominal marginal coverage assumes exchangeability, which chronological shift may violate. Finite-sample ranks beyond calibration size produce unbounded intervals; descriptive test coverage is not a future guarantee.

Training range/category indicators describe extrapolation and cannot estimate an OOD probability or guarantee validity.

Nonlinear structured branches and linear modalities share retained cohorts, but have different capacities. Additional complexity does not guarantee improved performance.

Tree impurity importances summarize global behavior, may favor high-cardinality features, and are neither trial-specific SHAP values nor causal evidence.

Text uses TF-IDF; BioBERT, survival, and SHAP are optional separate capabilities and are not claimed as executed.

Completed-only analysis cannot estimate censored/ongoing trial outcomes without a separate survival cohort.

Per-phase test metrics are exploratory; small or one-class subgroups may be unstable or undefined. Scatter plots retain at most 200 chronological rows; metrics and reliability bins use every retained row.

Research decision support only. Completed-only cohort creates follow-up/selection bias; minimum sample guard is not evidence of scientific validity. RETROSPECTIVE SNAPSHOT: baseline feature availability is unverified; post-start registry updates and narrative text may leak outcome information. Do not claim prospective validation.